Palindrome Reorder

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1755 | Introductory Problems | https://cses.fi/problemset/task/1755

Given a string, your task is to reorder its letters in such a way that it becomes a palindrome (i.e., it reads the same forwards and backwards).

Input

The only input line has a string of length n consisting of characters A–Z.

Output

Print a palindrome consisting of the characters of the original string. You may print any valid solution. If there are no solutions, print "NO SOLUTION".

Constraints

- $1 \le n \le 10^6$

Example

Input
AAAACACBA

Output
AACABACAA